5.5.15 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

April – May 2015

Max. Marks: 100 | **Duration: 3 Hrs**
Class: SE | **Semester: IV**
Name of the Course: Applied Mathematics-IV | **Branch: I.T /COMP**
Course Code: UITC401/UCEC401

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.		Questions	Marks
			20
Q.1	A	**Attempt any TWO**	12
	(i)	If $A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$ find, A^{100}	
	(ii)	If $A = \begin{bmatrix} 123 & 231 & 312 \\ 231 & 312 & 123 \\ 312 & 123 & 231 \end{bmatrix}$ & A^{-1} exists (i)Show that one of the Characteristic roots of A^3 is $(666)^3$ (ii) Show that one of the Characteristic roots of A^3 is negative.	
	(iii)	Is $A = \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix}$ derogatory?	
	B	**Attempt any ONE**	8
	(i)	Show that the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ is diagonalizable. Find the transforming matrix and the diagonal matrix	
	(ii)	If $A = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$ then Show that $e^A = e^3 \begin{bmatrix} cosh3 & sinh3 \\ sinh3 & cosh3 \end{bmatrix}$	
Q.2			20
	A	**Attempt any TWO**	12
	(i)	Evaluate $\int_c \tan z\, dz$ where $c: \lvert z\rvert = 2\pi$	

(ii) Evaluate $\int_c \frac{z^2}{((z-1)^2)(z+2)}\, dz$ where(i) c: $|z| = 0.5$ (ii) c: $|z| = 1.5$ (iii) c: $|z| = 2.5$

(iii) Evaluate $\int_0^{2\pi} \frac{\cos(3\theta)}{5+4\cos(\theta)}\, d\theta$

B Attempt any TWO — 8

(i) Expand $f(z) = \frac{z-1}{(z^2-2z-3)}$ within the unit circle about the origin using the Laurent's series

(ii) Evaluate $\int_c \frac{ze^z}{((z-2)^3)}\, dz$ where c: $|z| = 4$

(iii) Evaluate $\int_0^{1+i} (x - y + ix^2)\, dz$ along parabola $y^2 = x$

Q.3 — 20

A Attempt any TWO — 12

(i) A box contains three biased coins A,B & C. The probability that a head will result when the coin A is tossed is $\frac{1}{3}$, when the coin B is tossed is $\frac{2}{3}$ & when the coin C is tossed is $\frac{3}{4}$.If one of the coins is chosen at randon & is tossed 3 times . head resulted twice & tail once .What is the probability that the coin chosen was B .

(ii) The incomes of a group of 10000 persons were found to be normally distributed with mean Rs. 520 & S.D. Rs. 60. Find (i)the number of persons having incomes between Rs. 400 & Rs 500 (ii)the lowest income of the richest 500.

(iii) Fit a Poisson Distribution if

x	0	1	2	3	4
F	123	59	14	3	1

B Attempt any TWO — 8

(i) Find the M.G.F. of a random variable whose probability mass function is $P(X=x) = ({}^1/_2)^x$, x=1,2,3,......

(ii) A continuous random variable X has the p.d.f. defined by $f(x)=A+Bx$, $0 \le x \le 1$. If the mean of the distribution is $\frac{1}{3}$,find A&B

(iii) The ratio of the probability of 3 successes in 5independent trials to the probability of 2 successes in 5independent trials is $\frac{1}{4}$,find p&q

— 20

A Attempt any TWO — 12

(i) Find the coefficient of Rank correlation for the following data :

X	32	55	49	60	43	37	43	49	10	20
Y	40	30	70	20	30	50	72	60	45	25

	(ii)	The average marks scored by 32 boys is 72 with a S.D. of 8, while that for 36 girls is 70 with a S .D. of 6. Test at 1% level of significance whether the boys perform better than girls.	
	(iii)	Investigate the association between the darkness of eye colour in mother & son for the following data	

		Colour of mother's eyes		
		Dark	Notdark	Total
Colour of son's eyes	Dark	48	90	138
	Notdark	80	782	862
	Total	128	872	1000

	B	**Attempt any ONE**	8
	(i)	Find Karl Pearson's coefficient of correlation & hence obtain(i) equation of regression lines of x on y (ii) equation of regression lines of y on x	

X	28	45	40	38	35	33	40	32	36	33
Y	23	34	33	34	30	26	28	31	36	35

(ii) Two independent samples of sizes 8 &7 contained the following values

SampleI	19	17	15	21	16	18	16	14
SampleII	15	14	15	19	15	18	16	

Is the difference between sample means significant ?

Q.5			20
	A	**Attempt any TWO**	12

(i) Solve the following L.P.P. by Simplex method

maximise $z = 5x_1 + 4x_2$

Subject to $6x_1 + 4x_2 \leq 24$

$x_1 + 2x_2 \leq 6$

$-x_1 + x_2 \leq 1$

$x_2 \leq 2$

where $x_1 \geq 0, x_2 \geq 0$

(ii) Using the method of Lagrange's Multiplier solve the following N.L.P.P.

Optimize $z = 12x_1 + 8x_2 + 6x_3 - x_1^2 - x_2^2 - x_3^2 - 23$

Subject to $x_1 + x_2 + x_3 = 10$

where $x_1, x_2, x_3 \geq 0$

(iii) Find (i) All basic solutions (ii) All feasible basic solutions (iii) Optimal feasible basic solution for the following L.P.P.

maximise $z = 2x_1 - 2x_2 + 4x_3 - 5x_4$

Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$

$-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$

where $x_1, x_2, x_3, x_4 \geq 0$

	B	**Attempt any ONE(Mathematical Programming)**	8
	(I)	Using the Kuhn-Tucker conditions, solve the following N.L.P.P. Maximise $z = x_1^2 + x_2^2$ Subject to $x_1 + x_2 \leq 4$ $2x_1 + x_2 \leq 5$ where $x_1, x_2 \geq 0$	
	(ii)	Use the dual simplx method to solve the following L.P.P. minimise $z = 6x_1 + 3x_2 + 4x_3$ Subject to $x_1 + 6x_2 + x_3 = 10$ $2x_1 + 3x_2 + x_3 = 15$ where $x_1, x_2, x_3 \geq 0$	